# Giacomina (Mina) Grandolfo

*PhD Candidate in Cognitive Neuroscience | Université Libre de Bruxelles*

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## RESEARCH PROFILE

I investigate the cognitive and neural mechanisms that distinguish lucid from non-lucid dreamers during wakefulness, with a focus on interoception, metacognitive monitoring, and multisensory self-representation. My work is grounded in the predictive processing framework and uses signal detection theory, computational modelling (HMeta-d), and psychophysical paradigms. A core finding across experiments is that frequent lucid dreamers do not show superior raw interoceptive perception, but do show enhanced metacognitive sensitivity *(M-ratio advantage: $p = .033$, $d = 0.77$)* and greater resistance to multisensory body illusions at high interoceptive accuracy. suggesting that lucid dreaming reflects enhanced second-order monitoring of self-states rather than first-order perceptual precision.

## EDUCATION

**PhD in Cognitive Neuroscience (in progress)** *2023 to present*

Université Libre de Bruxelles (ULB) | CRCN | Supervisors: Axel Cleeremans & Peter Simor

Thesis: "Lucid Dreamers, Interoception and Multisensory Integration"

Intermediate examination passed with distinction, October 2025

**MSc Psychological Sciences. Research in Cognitive Sciences (cum laude)** *2021 to 2023*

Université Libre de Bruxelles (ULB)

**Computer Science (2 years completed). Software Development & Networking** *2020 to 2022*

IEPSCF Uccle, Brussels | Coursework in programming, database systems, networks, and systems administration

**MSc Labour Sciences** *2014 to 2016*

Université Libre de Bruxelles (ULB) | Erasmus exchange, Universidad de Sevilla

**Bachelor's degree in Special Education** *2011 to 2014*

HEB Defré, Brussels

## RESEARCH EXPERIENCE

**Doctoral Research. Lucid Dreaming, Interoception & Multisensory Integration** *2023 to present*

CRCN, Université Libre de Bruxelles

**Experiment 1 (completed, manuscript finalised, soon under review at Neuroscience of Consciousness)**

- Cardio-visual enfacement illusion ($n = 47$; 22 fLD, 25 nLD): frequent lucid dreamers showed higher interoceptive accuracy and, at high IAcc, reversed the illusion. suggesting a more rigid self-model boundary
- Key stats: IAcc group difference $t(45) = 2.61$, $p = .012$, $d = 0.77$; synchrony × IAcc × group interaction, $F = 4.06$, $p = .049$, $\eta p^2 = .084$

**Experiment 2. HRD Heartbeat Detection Task (ongoing, n = 53)**

- Interoceptive metacognition (M-ratio via HMeta-d): lucid dreamers show metacognitive advantage ($p = .033$, $d = 0.77$) and stronger calibration updating effect ($p = .026$, $d = 0.74$)
- Exteroceptive AUROC2: no significant group difference at current sample size. selective interoceptive effect
- Full R pipeline (scripts 01-05 + gen_ld_final.R); data collection and analysis ongoing

Ongoing collaboration with Audrey Mazancieux (Université Grenoble-Alpes). metacognition researcher who contributed to the R implementation of the m-ratio framework (metaSDT / HMeta-d workflow); joint literature review on metacognition and lucid dreaming planned for summer 2026.

**Experiment 3. Rubber Hand Illusion (ongoing, n > 55)**

- Proprioceptive drift and body ownership in lucid vs. non-lucid dreamers; R analysis pipeline active

**Experiment 4. VR Rod-and-Frame Task (data collection started April 2026, n = 14)**

- Visual-vestibular integration with Bayesian adaptive staircase (PSI); 2×2 design (head tilt × postural demand)
- Unity VR environment; Bayesian model fitting (Alberts et al., 2016 framework in Python)

## PUBLICATIONS & MANUSCRIPTS

**Grandolfo, G., Simor, P., & Cleeremans, A.** (under review). Lucid dreamers resist the cardio-visual enfacement illusion: interoceptive accuracy moderates multisensory self-other boundaries. *Neuroscience of Consciousness*

**Grandolfo, G.** (in preparation). Enhanced interoceptive metacognition in frequent lucid dreamers: evidence from the heartbeat discrimination task. Target journal: *Psychological Science*.

**Grandolfo, G. & Mazancieux, A. (in preparation). Metacognition and lucid dreaming: a literature review. Joint project with Audrey Mazancieux (Universite Grenoble-Alpes).**

## CONFERENCE PRESENTATIONS & SCIENTIFIC ACTIVITIES

**ASSC29. 29th Meeting of the Association for Scientific Study of Consciousness** *July 2026*

Santiago, Chile | Oral/Poster: "Lucid dreaming modulates the cardio-visual enfacement illusion" | Abstract accepted

**ESRS 2026. European Sleep Research Society** *2026*

Abstract submitted: Lucid dreaming, interoceptive precision and the enfacement illusion

**ASSC27. 27th Meeting, Tokyo | Poster presentation** *2024*

"Lucid dreaming and self-recognition: interoceptive signals modulate body ownership illusions"

**ASCC24. Association for the Scientific Study of Consciousness | Poster** *July 2024*

Participation and poster presentation, 2-5 July 2024

**Cognitive Control & Performance Monitoring (Wim Gevers, ULB) | Co-organiser** *June 2024*

3-day symposium. active participation in full organisation committee

**NeuroCog2023. Cognitive Neuroscience Symposium | Attendee** *Nov 2023*

**CuttingEEG 2025. Advanced EEG Methods Workshop | Attendee** *2025*

Hands-on training in MNE-Python, MEEG source analysis, and advanced EEG pipelines

**UNI General Meeting | Abstract presented** *2025*

"Lucid dreaming, interoception and multisensory integration: evidence from waking paradigms"

## TEACHING & SUPERVISION

**Intern Supervisor | CRCN, ULB** *2024 to 2026*

- Supervised 6 research interns across multiple cohorts (Cseniya, Chris, Jade, Karim, Joshua Nicdao Blanco, + 1)
- Responsibilities: weekly meetings, task assignment, experimental protocol training (HRD task, VR rod-and-frame), data quality monitoring, mid-stage and final evaluations

**Master's Thesis Supervision | ULB** *2025 to 2026*

Primary supervisor for one MSc thesis (Inès). two drafts reviewed and annotated

**Student Teaching Assistant. Neuropsychology | ULB** *2021*

Guidance sessions (2 h/week): concept reformulation, Wooclap-based active quizzing, Q&A support

**Student Delegate. Bachelor's Psychology | ULB** *2020*

Liaison between professors and students; member of the student bureau

## PROFESSIONAL EXPERIENCE

**AVS Elektro SPRL. Managing Director & Technical Project Manager** *2014 to 2017*

Brussels-based SME specialising in electrical engineering, home automation and security systems

- Directed technical projects: access control systems (badge readers, RFID, IP intercom), CCTV, home automation (KNX/Domotics) and audio distribution
- Produced AutoCAD technical cable drawings and wiring schematics for residential and commercial clients
- Managed project lifecycle: client intake, site survey, specifications, contractor coordination, commissioning and handover
- Human resources management, subcontractor relations, financial oversight

**Chargée de communication | Amaragea ASBL** *2011 to 2014*

First youth service in Brabant Wallon. Webmaster, content creation, project management

**Éducatrice | Centre Ferme Canada & Prison d'Everberg (minors)** *2009 to 2011*

Psycho-educational support in closed residential facilities; individual and group interventions

## TECHNICAL SKILLS

**Programming & Computational Tools**

Python. psychophysics experiment control (cardioception wrappers for HRD), scipy-based Bayesian model fitting, validation pipelines

R (advanced). tidyverse, BayesFactor, lme4, metaSDT; hierarchical Bayesian meta-d' (HMeta-d) workflow; full analysis pipelines for Experiments 1-3

C#. Unity VR development; adaptive Psi marginal method (Prins 2013 / Legrand 2022) implemented and validated against Python reference

MATLAB. basic familiarity (HMeta-d reference implementation)

**Statistical & Computational Methods**

Signal detection theory (d', meta-d', AUROC2)

Hierarchical Bayesian modelling of metacognition (HMeta-d)

Bayesian computational modelling: Alberts et al. (2016) visuo-vestibular integration model, 6-7 free parameters, SSE + MLE fitting

Adaptive psychophysical methods (staircase, Psi marginal)

Linear mixed models, Bayesian t-tests, ANOVAs

**Experimental & Hardware Platforms**

Unity VR development (HTC Vive Focus Vision; OpenXR)

BioSemi ActiveTwo ECG/EEG (planned Experiment 00)

MNE-Python for EEG analysis (CuttingEEG 2025 training)

Eye-tracking, heart rate acquisition (Polar, BioSemi)

**Languages**

French (native), Italian (native), English (fluent, scientific working proficiency), Spanish (intermediate, Erasmus Sevilla)

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| **Programming & Data** | R (tidyverse, afex, BayesFactor, metafor), Python (MNE, NumPy, SciPy, pandas, PsychoPy), MATLAB, AutoHotkey, Arduino (C/C++), LaTeX |
| **Statistics & Modelling** | Signal Detection Theory (SDT), HMeta-d (hierarchical Bayesian metacognitive efficiency), mixed-effects models, PSI adaptive staircase, Bayesian model fitting (MCMC/Stan), JASP |
| **EEG / Neuroscience** | BioSemi ActiveTwo, MNE-Python, EEGLAB, cardioception library, ECG/HEP acquisition (M5Stack), basic source analysis (CuttingEEG 2025) |

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| **Experimental platforms** | PsychoPy/PsychoJS, Unity (VR), SONA participant management, OSF pre-registration |
| **Technical / Engineering** | AutoCAD (2D wiring & cable plans), KNX/DALI home automation, access control (RFID/IP), CCTV installation, structured cabling |
| **Languages** | French (native), Italian (native), English (professional), Spanish (intermediate. Erasmus) |

*References available on request | April 2026*